

PHYSICS DT-3/8/23

- 21) Astigmatism
- 22) Repulsion
- 23) Laterally Inverted
- 24) Below the Image of Fish

25) The focal length of the lens = $\frac{1}{1.5} \text{ m}$ (as $P = \frac{1}{f} \text{ or } f = \frac{1}{P}$)

$$\text{So, } \frac{1}{1.5} \times \frac{10}{10} = \frac{10^2}{153} \Rightarrow \frac{2}{3} \text{ m} = 0.67 \text{ m}$$

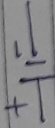
The lens is ^{Converging} diverging or a Conver lens.

26) We know $I = \frac{q}{t}$

given, $I = 5 \text{ A}$, $t = 30 \text{ minutes}$

$$\begin{aligned} q &= I \times t \\ &= 5 \times 30 \times 60 \text{ C} \\ &= 150 \times 60 \text{ C} \\ &= 9000 \text{ C} \end{aligned}$$

27) Circuit symbol of five important components are:-

- 1)  $\rightarrow$ Cell
- 2)  $\rightarrow$ Bulb
- 3)  $\rightarrow$ Resistor
- 4)  $\rightarrow$ Key
- 5)  $\rightarrow$ Switch (OFF)
- 6)  $\rightarrow$ Switch (ON)
- 7)  $\rightarrow$ Ammeter
- 8)  $\rightarrow$ Voltmeter

3)

Given, lens nature is convex

Focal length = +12cm

 $h_o = 5\text{cm}$

Object distance = -20cm

Using lens formula,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\text{or, } \frac{1}{v} - \frac{1}{-20} = \frac{1}{12}$$

$$\text{or, } \frac{1}{v} + \frac{1}{20} = \frac{1}{12}$$

or,

$$\frac{1}{v} = \frac{1}{12} - \frac{1}{20}$$

$$\text{or, } \frac{1}{v} = \frac{5-3}{60} = \frac{2}{60}\text{cm}$$

$$\text{or } v = \frac{60 \times 30}{2} \Rightarrow \boxed{90\text{cm}} \rightarrow \text{Image distance.}$$

Size of Image $\rightarrow$

$$\frac{v}{u} = m \text{ (magnification)}$$

$$\text{or, } \frac{90}{-20} \text{cm}$$

$$\text{or } \frac{-9}{2}$$

$$-\frac{3}{2} \times \frac{h_i}{h_o} \Rightarrow \frac{-3}{2} \times \frac{h_i}{5\text{cm}} \Rightarrow \frac{-15}{2} \times \frac{h_i}{5\text{cm}} \Rightarrow \frac{-7.5\text{cm}}{1}$$

Hence, the nature of image is real, inverted and magnified.

2) The charge is to be moved from higher potential to lower potential, i.e. $10V \rightarrow 5V$.

$$V_B - V_A = \frac{W_{AB}}{q_0}$$

$$\text{or, } \frac{5V - 10V}{1} = \frac{5J}{q_0}$$

$$\text{or, } -5V = \frac{5J}{q_0}$$

$$\text{or, } 5q_0 = 5J$$

$$\text{or, } q_0 = \frac{5J}{5V} = 1C$$

PART I

1) Pole - It is a geometrical centre of the reflecting surface. It is represented by letter 'P'.

Radius of curvature - The radius of the sphere of which the mirror is a part of.

Principal Axis - The line joining the pole and Centre of curvature of the mirror.

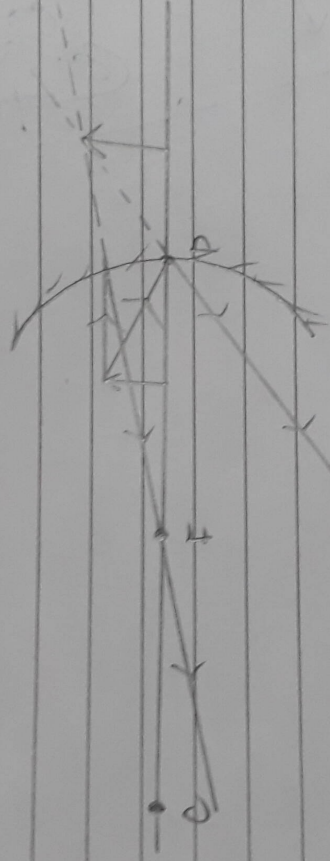
Principal focus - It is a point on the principal axis where at which rays parallel to principal axis meet after reflection.

PART II → The range of distance is between the pole and the focus, i.e. $0cm$ to $20cm$.

Characteristics of observed image

- i) Magnified image
- ii) Formed behind the mirror.
- iii) Virtual image.

RAY DIAGRAM



Concave mirror

- i) The sun appears reddish in the early morning as it is closer to the horizon and undergoes atmospheric refraction.
- ii) Power of accommodation of human eye means the ciliary muscle would focus the image on retina when the image is far as near.
- iii) The least distance of distinct vision of human eye is 25cm.